

Item ID	Category	Component Name	Specification / Model	#	Quantity	🔗	Phase	Function / Application
PWR-01	Solar Panel	Photovoltaic Panel	5W / 6V Polycrystalline	1	1		Phase 1	Static solar energy harvesting for Pocket MVP
PWR-02	Solar Panel	Photovoltaic Panel	30W / 18V Polycrystalline	1	1		Phase 3	Main solar harvesting for Off-Grid Power Hub
PWR-03	Power Management	Charge Controller Module	TP4056 with Protection Circuit (1A)	1	1		Phase 1	Single-cell (1S) Li-ion charge & protection manager
PWR-04	Power Management	Boost Converter Module	Step-Up 5V Regulated USB Output	1	1		Phase 1	Voltage step-up to regulated 5V USB output
PWR-05	Power Management	BMS Board	3S 40A with Charge Balancing	1	1		Phase 3	Overcharge overdischarge and balancing for 12V battery
PWR-06	Power Management	Step-Down Buck Regulator	Multi-Port 9V-36V Input (2x USB-C PD + 1x QC3.0)	1	1		Phase 3	High-efficiency step-down conversion for fast charging
BAT-01	Energy Storage	18650 Li-ion Battery Cells	High-Capacity 18650 (LG / Samsung / Panasonic)	6	6		Phase 1 & 3	Energy storage buffer (1 cell in Phase 1 3S2P in Phase 3)
BAT-02	Accessories	Battery Holder Slot	18650 Battery Holder Slot	1	1		Phase 1 & 3	Mechanical enclosure and secure electrical connection
MCU-01	Automation	Microcontroller Board	Arduino Uno R3 / Nano V3 + USB Cable	1	1		Phase 2	Tracking algorithm processing & PWM motor control
MCU-02	Automation	Breadboard	830-Point Solderless Breadboard	1	1		Phase 2	Prototyping signal and control circuits
MCU-03	Automation	Jumper Wires	Male-to-Male & Male-to-Female Wire Pack	1	1		Phase 2	Circuit signal routing and power distribution
SEN-01	Sensors	LDR Photoresistor	5mm LDR Light Sensor	4	4		Phase 2	4-quadrant solar irradiance sensing matrix
RES-01	Passive Components	Precision Resistor	10 kΩ (1/4W)	4	4		Phase 2	Voltage divider network for LDR analog reading
RES-02	Passive Components	Current Limiting Resistor	220 Ω (1/4W)	10	10		Phase 2	Current limiting for LED status indicators
ACT-01	Actuators	Mini Servo Motor	SG90 (9g)	1	1		Phase 2	Low-torque proof-of-concept movement validation
ACT-02	Actuators	High-Torque Servo Motor	MG996R (Metal Gears)	2	2		Phase 2	Dual-axis actuation (Azimuth / Horizontal & Elevation)